Grokking Microservices Design Patterns

Master microservices design patterns for designing scalable, resilient, and more manageable systems.

Level: Intermediate | Study Time: 60h | Lessons: 93 | Playgrounds : 1

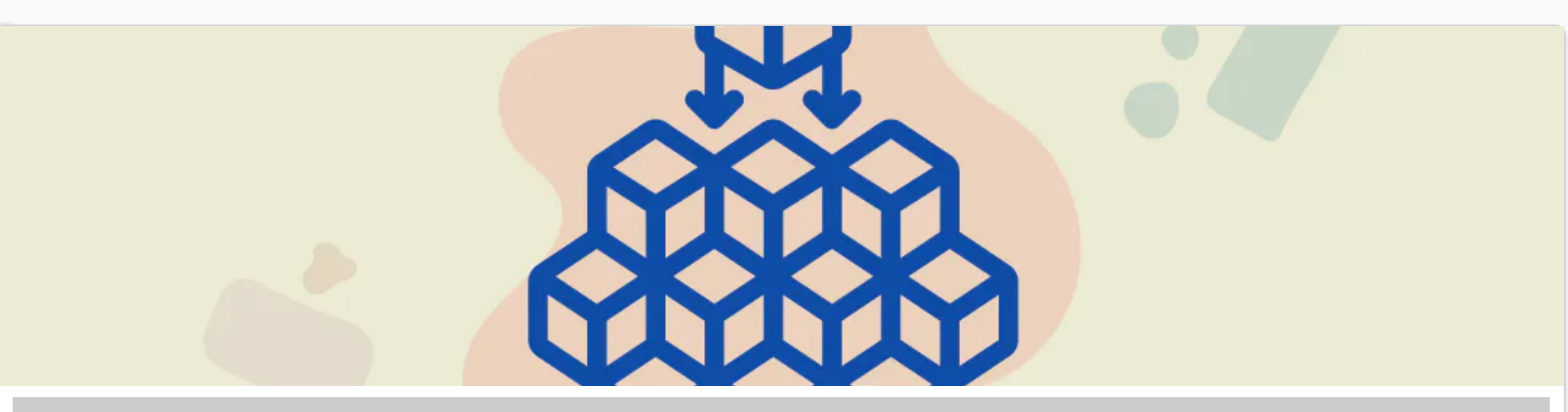
Course Overview

Delve into the world of microservices with this comprehensive online course, designed to equip software professionals with the tools they need to tackle modern software architecture. Through this course, learners will gain a deep understanding of the principles, patterns, and best practices that underpin microservice architectures.

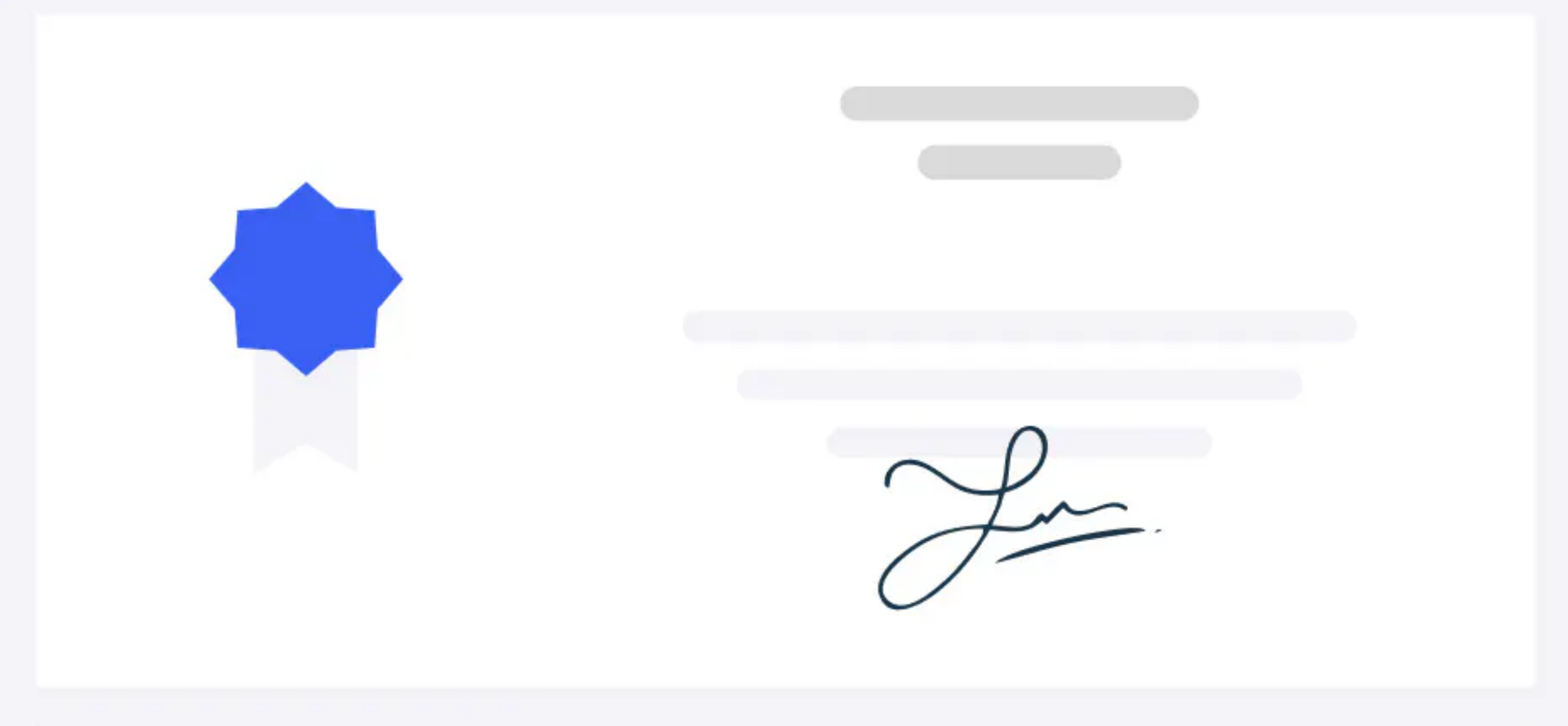
This course sheds light on why microservices have become the go-to architecture for scalable and resilient applications, from startups to tech giants. It navigates through the essential design patterns like Sidecar, API Gateway, Circuit Breaker, and more, providing real-world examples and code snippets.

Whether you're a seasoned developer, a budding software architect, or an IT professional looking to stay ahead of the curve, this course offers valuable insights into how to design and manage complex systems with microservices. Immerse yourself in this fascinating journey and embrace the future of software architecture.

Join us in this course and gain the skills and knowledge you need to thrive in the ever-evolving world of technology. Your journey towards mastering microservice design patterns starts here.



Start Reading



Certification on completing the course.

Download Certificate

Course Content

1. Introduction

2 chapters

- ✓ Who Should Take This Course?
- ✓ The Course at a Glance

2. Strangler Fig Pattern

6 chapters

- ✓ Introduction
- ✓ The Problem: Legacy Systems
- ✓ The Strangler Pattern: A Solution
- ✓ The Architecture of the Strangler Pattern
- ✓ Strangler Pattern: A Detailed Example
- ✓ Key Insights and Implications

3. API Gateway Pattern

5 chapters

- ✓ Introduction to the API Gateway Pattern
- ✓ Advantages of API Gateway Pattern
- ✓ API Gateway Pattern: An Example
- ✓ Performance Implications
- ✓ System Design Example

4. Backends for Frontends (BFF) Pattern

6 chapters

- ✓ Introduction to BFF
- ✓ The Problem: Traditional Backend Models
- ✓ The Architecture of the BFF Pattern
- ✓ BFF Pattern: An Example
- ✓ Performance Implications
- ✓ System Design Examples

5. Service Discovery Pattern

8 chapters

- ✓ What is Service Discovery Pattern?
- ✓ The Problem: Service Coordination in Distributed Systems
- ✓ Service Discovery Pattern: A Solution
- ✓ The Architecture of the Service Discovery Pattern
- ✓ The Inner Workings of the Service Discovery Pattern
- ✓ Service Discovery Pattern: An Example
- ✓ Performance Implications and Special Considerations
- ✓ System Design Examples

6. Circuit Breaker Pattern

6 chapters

- ✓ Introduction
- ✓ The Problem: The Struggles of Distributed Systems and Service Failures
- ✓ The Circuit Breaker Pattern: An Effective Shield Against Cascading Failures
- ✓ Circuit Breaker Pattern: An Example
- ✓ Performance Implications and Special Considerations
- ✓ System Design Examples

7. Bulkhead Pattern

9 chapters

- ✓ Introduction
- ✓ The Problem: Failure Propagation in Distributed Systems
- ✓ The Bulkhead Pattern: A Solution
- ✓ The Architecture
- ✓ The Inner Workings
- ✓ Bulkhead Pattern: A Example
- ✓ Performance Implications and Special Considerations
- ✓ System Design Examples
- ✓ Conclusion

8. Retry Pattern

9 chapters

- ✓ Introduction
- ✓ The Problem: Unreliable External Resources in Distributed Systems
- ✓ The Retry Pattern: A Solution
- ✓ The Architecture of the Retry Pattern
- ✓ The Inner Workings of the Retry Pattern
- ✓ Retry Pattern: An Example
- ✓ Performance Implications
- ✓ Use Cases and System Design Examples
- ✓ Conclusion

9. Sidecar Pattern

7 chapters

- ✓ Introduction to the Sidecar Pattern
- ✓ The Problem: Monolithic Application Management
- ✓ A Solution to the Monolithic Mayhem
- ✓ The Architecture of the Sidecar Pattern
- ✓ Sidecar Pattern: Bringing Theory to Practice with an Example
- ✓ Performance Implications
- ✓ System Design Examples: Bringing the Sidecar Pattern to Life

10. Saga Pattern

9 chapters

- ✓ Introduction to Saga Pattern
- ✓ The Problem: Traditional Transaction Models
- ✓ The Saga Pattern: A Solution
- ✓ The Architecture of the Saga Pattern
- ✓ The Inner Workings of the Saga Pattern
- ✓ Saga Pattern: A Example
- ✓ Performance Implications
- ✓ System Design Examples
- ✓ Conclusion

11. Event-Driven Architecture Pattern

9 chapters

- ✓ Introduction
- ✓ The Problem: Managing Complex Interactions in Distributed Systems
- ✓ Event-Driven Architecture: A Promising Solution
- ✓ The Architecture of the Event-Driven Architecture Pattern
- ✓ The Inner Workings of the Event-Driven Architecture Pattern
- ✓ Event-Driven Architecture Pattern: An Example
- ✓ Performance Implications and Special Considerations
- ✓ Use Cases and System Design Examples
- ✓ Conclusion

12. CQRS (Command Query Responsibility Segregation)

8 chapters

- ✓ Introduction
- ✓ The Problem: Traditional CRUD Operations
- ✓ CQRS Pattern: A Solution
- ✓ The Architecture of the CQRS Pattern
- ✓ The Inner Workings of the CQRS Pattern
- ✓ CQRS Pattern: An Example
- ✓ Issues, Special Considerations, and Performance Implications
- ✓ System Design Examples

13. Configuration Externalization Pattern

8 chapters

- ✓ Introduction
- ✓ The Problem: Configuration Management in a Microservices Architecture
- ✓ The Solution: Configuration Externalization Pattern
- ✓ Unveiling the Architecture: How Does Configuration Externalization Work?
- ✓ Delving into Code: An Example
- ✓ Considerations and Implications
- ✓ Use Cases and Real-world Examples
- ✓ Conclusion

14. Course Wrap-up

1 chapter

- Embrace the Future of Software Architecture